Loan Default Risk Analysis (SQL findings)

📃 Project Overview

This project analyzes loan data using SQL to identify **default patterns, risk factors, and revenue exposure**. The goal is to help financial institutions make **better lending decisions** and reduce risk.

Business Problem 1 — Overall Portfolio Health

SQL Query

```
SELECT

    COUNT(*) AS total_loans,

    SUM(loan_status) AS total_defaults,

    ROUND(AVG(loan_status) * 100, 2) AS default_rate_pct,

    ROUND(AVG(loan_amnt), 0) AS avg_loan_amount,

    ROUND(SUM(CASE WHEN loan_status = 1

              THEN loan_amnt ELSE 0 END), 0) AS total_amount_at_risk

FROM loan_data;
```

Findings

- Around **20–22% of loans are defaulting**
- This means **1 out of every 5 loans goes bad**
- A significant portion of total loan amount is **at financial risk**

Insight

The bank is exposed to **high default risk**, indicating the need for stricter approval policies.

Business Problem 2 — Risk by Loan Grade

SQL Query

```
SELECT

    loan_grade,

    COUNT(*) AS total_loans,

    SUM(loan_status) AS defaults,

    ROUND(AVG(loan_status) * 100, 2) AS default_rate_pct,

    ROUND(AVG(loan_int_rate), 2) AS avg_interest_rate,

    ROUND(AVG(loan_amnt), 0) AS avg_loan_amount

FROM loan_data

GROUP BY loan_grade

ORDER BY loan_grade;
```

Findings

- Grade A loans have **very low default rates (~5%)**
- Grade G loans have **extremely high default rates (~70%)**

Insight

Higher loan grades (F, G) are **unprofitable due to high defaults**

Recommendation

Avoid approving high-risk grades or increase interest rates to compensate.

Business Problem 3 — Income vs Default Risk

SQL Query

```
SELECT

    CASE
```

```
        WHEN person_income < 20000  THEN 'Below 20k'
        WHEN person_income < 40000  THEN '20k-40k'
        WHEN person_income < 60000  THEN '40k-60k'
        WHEN person_income < 80000  THEN '60k-80k'
        WHEN person_income < 100000 THEN '80k-100k'
        ELSE 'Above 100k'
    END AS income_bracket,
    COUNT(*) AS total_loans,
    ROUND(AVG(loan_status) * 100, 2) AS default_rate_pct,
    ROUND(AVG(loan_amnt), 0) AS avg_loan_amount
FROM loan_data
GROUP BY income_bracket
ORDER BY MIN(person_income);
```

Findings

- Low-income group (<20k) → **high default (~35–40%)**
- High-income group (>100k) → **low default (~5%)**

Insight

Income is a **strong predictor of loan repayment ability**

Recommendation

Apply stricter checks for low-income applicants.

Business Problem 4 — Employment Stability

SQL Query

```
SELECT
    CASE
        WHEN person_emp_length IS NULL THEN 'Unknown'
```

```
        WHEN person_emp_length < 1 THEN 'Less than 1 yr'
        WHEN person_emp_length < 3 THEN '1-3 years'
        WHEN person_emp_length < 5 THEN '3-5 years'
        WHEN person_emp_length < 10 THEN '5-10 years'
        ELSE '10+ years'
    END AS employment_band,
    COUNT(*) AS total_loans,
    ROUND(AVG(loan_status) * 100, 2) AS default_rate_pct
FROM loan_data
GROUP BY employment_band
ORDER BY MIN(COALESCE(person_emp_length, -1));
```

Findings

- Customers with <1 year experience → **high default**
- Customers with 10+ years → **low default**
- Unknown employment → risky segment

Insight

Job stability directly impacts repayment behavior

Recommendation

Make employment history mandatory and flag unstable profiles.

Business Problem 5 — Loan-to-Income Ratio

SQL Query

```
SELECT
    CASE
        WHEN loan_percent_income < 0.10 THEN 'Under 10%'
```

```
        WHEN loan_percent_income < 0.20 THEN '10%-20%'
        WHEN loan_percent_income < 0.30 THEN '20%-30%'
        WHEN loan_percent_income < 0.40 THEN '30%-40%'
        WHEN loan_percent_income < 0.50 THEN '40%-50%'
        ELSE 'Over 50%'
    END AS loan_to_income_band,
    COUNT(*) AS total_loans,
    ROUND(AVG(loan_status) * 100, 2) AS default_rate_pct,
    ROUND(AVG(loan_amnt), 0) AS avg_loan_amount
FROM loan_data
GROUP BY loan_to_income_band
ORDER BY MIN(loan_percent_income);
```

Findings

- <20% ratio → safe loans
- 40–50% ratio → very high default risk

Insight

Customers borrowing beyond their capacity are **more likely to default**

Recommendation

Set maximum loan-to-income threshold (~40%)

Business Problem 6 — Past Default Behavior

SQL Query

```
SELECT
    cb_person_default_on_file AS prior_default,
    COUNT(*) AS total_loans,
```

```
    SUM(loan_status) AS current_defaults,

    ROUND(AVG(loan_status) * 100, 2) AS default_rate_pct,

    ROUND(AVG(cb_person_cred_hist_length), 1) AS avg_credit_history_yrs

FROM loan_data

GROUP BY cb_person_default_on_file;
```

Findings

- Prior default (Y) → ~40% default rate
- No history (N) → ~18% default rate

Insight

Past default is a **strong predictor of future default**

Recommendation

Use credit history as a key approval factor.

Final Conclusion

- Default risk is driven by:
- Low income
- High loan burden
- Poor credit history
- Unstable employment
- High-risk loan grades

By applying these insights, banks can:

- Reduce default rates
- Improve profitability
- Make smarter lending decisions